SAMPLE

History of Astronomy: The Space Age

Sputnik to Apollo

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Introduction

On the fourth of October, 1957, a radio pulse from a metal sphere the size of a beach ball crossed the sky every ninety-six minutes, and the world below it changed. The sphere was Sputnik 1, the first artificial satellite, and its steady beep marked the beginning of the Space Age: the period in which humanity's instruments and its people left the planet not briefly, on the arcs of ballistic rockets, but to stay, orbiting and flying onward and, within twelve years, standing on the Moon. This book is the history of that era's opening arc, from Sputnik to Apollo, told as the History of Astronomy series tells it: as a story about what knowing the universe came to mean once machines could go there.

The Space Age did not begin with a discovery in the usual sense. Nothing in nature was found on that October evening. What changed was a demonstrated capability and, flowing from it, a set of possibilities that had belonged to speculation since Kepler wrote about voyaging to the Moon. Once one satellite orbited, the laws of mechanics guaranteed that others could, and that they could look downward at the whole Earth, outward past an atmosphere that had filtered starlight since the first eye opened, and eventually carry human beings with them. Astronomy, which had spent its entire history observing from the bottom of an ocean of air, gained a mountaintop above the ocean. The satellite era, the crewed era, and the era of instruments in space are one continuous development, and it started at a moment anyone with a shortwave receiver could hear.

The story has a rivalry at its center, and no honest account leaves that out. The Soviet Union launched the first satellite, the first animal, the first human, the first spacewalker, the first probe to hit the Moon, and the first photographs of the Moon's far side. The United States absorbed those shocks, reorganized its institutions, and spent its way to the Moon landings, which remain the era's defining human achievement. But the rivalry is not the whole story. The era also built weather satellites, communication satellites, navigation satellites, and a growing habit of cooperation that outlived the Cold War which spawned it. The tension between competition and shared enterprise runs through every chapter that follows.

The chapters ahead follow the series pattern. They open with the discovery of the era and the understanding of what it was, move through the science and the machinery that made it work, place the era in its Cold War and cultural context, and then turn to what the Space Age actually returned: the discoveries, the challenges and tragedies, the paths research took afterward, and the small human facts that official histories tend to lose. Because this is a history of astronomy, the science chapters carry the argument. The Space Age changed astronomy more than any development since the telescope, and the reasons are concrete: instruments above the atmosphere, probes at the planets, and a species that finally saw its own planet whole.

A note on how the account is built. The book works from the established public record of the era: dates, missions, hardware, and outcomes that are documented beyond dispute. Where a famous story exists in competing versions, the text says so rather than choosing the better anecdote. Where an achievement was collective rather than individual, it is described that way. The Space Age was the work of hundreds of thousands of people over decades, and it deserves the accuracy it did not always receive in its own publicity. The era began with a beep, but it was built with patience, and it is best understood the same way.

Chapter 1: The Discovery of The Space Age

No single person discovered the Space Age, but its beginning can be dated to the minute. At 1928 Greenwich Mean Time on 4 October 1957, the Soviet Union's R-7 rocket lifted from the Tyuratam range in Kazakhstan and placed Sputnik 1, a polished sphere of aluminum alloy just under sixty centimeters across, into an elliptical orbit reaching more than nine hundred kilometers above the Earth. The launch had been planned for the International Geophysical Year, the eighteen-month program of coordinated Earth science running from July 1957 through 1958, in which sixty-seven nations had agreed to study the planet together. The satellite was a modest science payload, two radio transmitters whose beep was meant to be tracked by stations on the ground. Its scientific payload, however, mattered less than its existence. Something built by human hands was in orbit, and it stayed there until January, circling the planet more than a thousand times before friction pulled it home.

The technical lineage ran back through the war. The German V-2 rockets of 1944 were the first human objects to cross the edge of space on ballistic arcs, and after the war their captured hardware and engineers seeded the missile programs on both sides of the Cold War. At White Sands in New Mexico, V-2s launched vertically carried cameras that returned the first photographs of the Earth from space in 1946, curving horizon visible above the New

Mexico desert. Through the late 1940s and early 1950s, sounding rockets in both the United States and the Soviet Union carried instruments briefly past the top of the atmosphere, and a few researchers, James Van Allen chief among them, made careers of retrieving data from these minutes-long flights. These flights were exploration but not presence. A rocket goes up and comes back. An orbit is a decision to stay.

Orbit required a launcher capable of about eight kilometers per second, and the only machines close were the new intercontinental ballistic missiles. In July 1955 the United States announced it would attempt a satellite during the Geophysical Year, choosing the deliberately civilian Vanguard project, a slim research rocket, as its vehicle. Days later the Soviet Union announced it would do the same. The American program proceeded openly and slipped; the Soviet program, run in secret under the chief designer Sergei Korolev, was ready first. Korolev had pressed for a simple satellite that could fly quickly, and he got his way over the objections of colleagues who wanted more elaborate science. The R-7 missile that carried Sputnik aloft had first flown successfully in August 1957, a month and a half before the satellite launch, as a weapon. The Space Age was born from a sword, and everyone involved knew it.

The reaction on the ground was immediate and, in the United States, close to panic. Amateur observers with binoculars and shortwave radios tracked the satellite and its spent booster, a bright tumbling cylinder that made more visible passes than the small sphere itself. Newspapers ran orbital predictions alongside freight rates. The beep at 20.005 megahertz was heard by ham operators worldwide, a sound at once trivial and historic, and in the weeks that followed the Soviet Union granted the sort of triumph that state propaganda could never have purchased. The United States attempted its Vanguard launch in early December before television cameras, and the rocket rose a meter or two, settled back, and exploded. The failure was christened with the nicknames it deserved, and the gap between announcement and capability became a political wound.

Sputnik's own science turned out to be modest but real. Tracking its radio through Doppler shifts told researchers about the density of the upper atmosphere, and the drag that shortened its orbit gave the same answer from another direction: the thin air at two hundred kilometers, while barely there, still counts. The Soviet Union followed on 3 November with Sputnik 2, a capsule of more than five hundred kilograms carrying the dog Laika, the first living creature sent into orbit, and a year later the United States finally succeeded, launching Explorer 1 on 31 January 1958, a small instrumented satellite that discovered a belt of charged particles ringing the Earth. Between the two programs the shape of the new era was set: propaganda value and scientific return in one package, and prestige measured in orbits. Within a year of the first beep, the United States had created a civilian space agency, and the race the decade had been waiting for had begun in earnest.

What had been discovered, in the end, was not a fact but a frontier that behaved differently from all previous frontiers. It cost enormous sums, it could not be held or fenced, and it rewarded engineering discipline more than daring alone. The next fifteen years would be spent finding out what that frontier could hold, and the price of learning was higher than the celebrations of 1957 and 1958 suggested.